

DyLogNet: A Dynamic Multi-Relational Graph Framework for Log Anomaly Detection

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Abstract

Web-scale platforms and online services rely on log-based anomaly detection to safeguard availability, latency SLOs, and user experience. In real-world web interactions, system logs often exhibit irregular temporal intervals, bursty densities, and heterogeneous semantics, which pose significant challenges for log anomaly detection. Existing methods such as LSTM and Transformer assume a fixed input window, which conflicts with the inherently irregular nature of system logs. Moreover, most prior works build a single-view representation, overlooking the multi-relational nature of logs. To overcome these challenges, we propose DyLogNet, a dynamic multi-relational graph framework for log anomaly detection. Specifically, this framework constructs a density-aware dynamic graph with variable-length windows, and represents logs from three relational perspectives: temporal co-occurrence, semantic similarity, and anomaly tendency. Next, we design a cross-layer attention mechanism that integrates heterogeneous structures to highlight the most relevant relations and enhance event representations. Furthermore, a cross-snapshot memory injection module updates global memory through a recurrent unit and injects it into current graph representations via an affine transformation, enabling temporal continuity. Experiments on three public log datasets demonstrate that DyLogNet outperforms state-of-the-art methods, especially in few-shot scenarios.

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CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; **Lexical semantics**; • **Software and its engineering** → **Software notations and tools**.

Keywords

Log analysis, Anomaly detection, Dynamic graph, Large Language Models

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1 Introduction

With the rapid proliferation of cloud-native architectures and web-scale platforms, modern online services have become increasingly complex, distributed, and dynamic [31, 38]. These systems comprise thousands of microservices, interdependent components, and asynchronous user interactions that continuously generate massive volumes of operational logs [23]. Such logs capture fine-grained runtime behaviors, configuration changes, and error traces, serving as a fundamental basis for reliability assurance, performance diagnosis, and automated operations [1, 16]. As the backbone of Artificial Intelligence for IT Operations (AIOps) pipelines, log analysis has thus become indispensable for maintaining service robustness and minimizing business disruptions [31].

At the same time, the unprecedented scale and heterogeneity of modern web systems have also made them more vulnerable to elusive and cascading anomalies [34, 39]. A single failure in one subsystem can quickly propagate across service dependencies, causing latency degradation, SLO violations, or even large-scale

outages [14, 24]. Consequently, timely and accurate log-based anomaly detection has emerged as a critical component of intelligent observability and web reliability engineering [35].

Traditional log anomaly detection frameworks typically consist of three key stages: (1) log parsing, which converts unstructured logs into structured event templates; (2) event embedding, which represents logs as numerical vectors capturing textual semantics and temporal patterns; and (3) model construction, which learns normal dependencies and identifies deviations from expected behaviors [8, 34]. Despite significant progress, existing approaches remain inadequate for modeling the intricate dynamics of real-world web logs [2, 29]. Sequence-based models such as LSTM and Transformer assume fixed temporal intervals and uniform sequence lengths, which contradict the inherently irregular and bursty nature of system logs [7]. Moreover, most methods focus on single-view sequence modeling, neglecting the rich multi-relational structures embedded in temporal co-occurrence, semantic affinity, and anomaly correlations [6, 33]. As a result, they struggle to capture higher-order dependencies and evolving contextual shifts that frequently arise in large-scale distributed systems [30].

These limitations call for a unified framework that can jointly model heterogeneous semantics, dynamic structures, and temporal evolution in logs. To this end, we propose **DyLogNet**, a dynamic multi-relational graph framework for web-scale log anomaly detection. DyLogNet strengthens log representations by integrating textual, numerical, and temporal signals into unified event embeddings. Building on these embeddings, it constructs a density-aware dynamic graph that adaptively captures dependencies through three complementary relation layers, reflecting temporal co-occurrence, semantic similarity, and anomaly propensity. A cross-layer attention module then fuses these heterogeneous relations and highlights the most informative structural cues. Moreover, to preserve long-term temporal continuity, DyLogNet incorporates a cross-snapshot memory injection module that recurrently updates and injects global memory into the evolving graph representations via a learnable affine transformation. Together, these components allow DyLogNet to adapt to irregular log streams and detect evolving anomalies with improved robustness and interpretability.

Extensive experiments on three public benchmark datasets demonstrate that DyLogNet consistently outperforms state-of-the-art methods in both accuracy and stability, particularly under few-shot and unstable logging conditions. Beyond performance improvements, DyLogNet provides a principled perspective for modeling web-scale logs as dynamic relational systems, paving the way toward more adaptive and trustworthy AIOps.

Contributions. The main contributions of this work are summarized as follows:

- We propose DyLogNet, a unified dynamic multi-relational graph framework for web-scale log anomaly detection that integrates semantic enrichment, relational graph modeling, and temporal continuity learning.
- We design a density-aware graph construction strategy and a cross-layer attention mechanism to jointly model temporal, semantic, and anomaly-specific relationships within evolving log streams.

- We develop a cross-snapshot memory injection module that maintains long-term temporal coherence and enables adaptive knowledge transfer across dynamic graph snapshots.
- Extensive experiments on three public benchmarks show that DyLogNet attains state-of-the-art performance and robust generalization, particularly in few-shot settings and under unstable logging conditions.

2 Related Work

2.1 Sequence-Based Anomaly Detection

As logs in modern software systems grow in scale and complexity, sequence-based deep learning has become the dominant approach for log anomaly detection [25]. By modeling logs as natural language sequences, these methods use architectures like RNNs and Transformers to capture temporal dependencies and semantic patterns. Despite their strengths in contextual modeling and handling pattern variability, they remain limited in embedding strategies, context-aware design, and robustness [29].

NeuralLog [2] avoids log parsing by encoding raw log text into semantic vectors and using a Transformer classifier to model contextual semantics for effective anomaly detection. LogAnomaly [21] combines template2vec embeddings with sequential and quantitative modeling to detect both sequence and numerical anomalies while reducing false alarms from unseen templates. LogBERT [9] employs a BERT-based framework with self-supervised tasks to learn contextual patterns in normal logs, enabling the detection of complex anomalies as pattern deviations.

Sequence-based approaches have advanced log anomaly detection, especially in unstructured and evolving log environments, but remain constrained by reliance on strict temporal order and limited context. To address these limitations, DyLogNet integrates graph-based representations with contextual enhancement to better capture multi-dimensional log relationships.

2.2 Graph-Based Anomaly Detection

Recently, Graph Neural Networks have shown great promise in log-based anomaly detection by effectively modeling complex nonlinear relationships, heterogeneous dependencies, and cross-temporal semantics. Unlike sequence-based models, graph-based approaches construct log graphs to explicitly capture structural and contextual information, enabling more robust and interpretable anomaly detection, especially in distributed or heterogeneous environments.

LogGD [26] converts log sequences into graphs and applies a Graph Transformer Network to fuse structural signals with node semantics, improving structural pattern modeling and mitigating false alarms when sequential cues are incomplete. HiLog [12] introduces a human-in-the-loop framework that combines expert knowledge with graph-based representations to detect four common anti-patterns in low-quality industrial logs. It improves detection accuracy under limited labels, demonstrating the benefit of incorporating human priors into graph models. HGAN [10] extends the heterogeneous graph modeling paradigm by combining node-level and semantic-level attention to learn the importance of nodes under different semantic contexts, achieving semantic-enhanced graph representations with superior performance and interpretability. Lograph [3] constructs a Log-Entity Graph through semantic

association mining and employs a Heterogeneous Graph Attention Network to effectively detect anomalies in complex, interleaved log environments without relying on predefined grouping strategies.

Compared with Lograph, DyLogNet better captures temporal dynamics and multi-faceted anomaly patterns. Lograph models interleaved logs with a Log-Entity Graph and heterogeneous attention but is less adaptive to evolving behaviors. DyLogNet instead leverages dynamic snapshots with adaptive windows and temporal decay, and a three-relation graph for temporal, semantic, and anomaly-oriented dependencies. With cross-layer attention and contrastive learning, it yields higher robustness and accuracy in dynamic logging environments.

3 DyLogNet

3.1 Overview

As shown in Figure 1, DyLogNet models irregular log streams by integrating semantic event encoding, density-aware snapshot graphs, multi-relation reasoning, two-stage attention fusion, and cross-snapshot memory. It embeds textual and numerical features into event representations, constructs temporally decayed snapshots with adaptive windows to capture temporal dependencies, semantic similarities, and anomaly-related relations, and fuses these views via attention. A recurrent memory with learnable affine injection maintains continuity across snapshots, while a lightweight temporal module and prediction head output anomaly predictions optimized with cross-entropy and contrastive objectives.

3.2 Enhanced Semantic Embeddings

The enhanced semantic embedding module aims to extract deep semantic features from log events, integrate multi-modal characteristics, and optimize embeddings for anomaly detection. This is achieved by leveraging pre-trained language models for textual encoding, mapping numerical features into the same latent space, and fusing both modalities through a gating mechanism.

Log events are first encoded using a pre-trained language model, such as BERT [4], to capture contextual and semantic information. Given a log event template or textual content x_t^{text} , its semantic embedding $e_t^{\text{text}} \in \mathbb{R}^d$ is obtained as:

$$e_t^{\text{text}} = \text{BERT}(x_t^{\text{text}}), \quad (1)$$

where d is the embedding dimension. To complement textual features, numerical attributes are mapped into the same embedding space through a non-linear transformation:

$$e_t^{\text{num}} = \text{MLP}(x_t^{\text{num}}) \in \mathbb{R}^d, \quad (2)$$

where x_t^{num} denotes the numerical feature vector and MLP represents a multi-layer perceptron.

To obtain a unified representation, the two embeddings are combined using a gated fusion mechanism:

$$g_t = \sigma(W_g[e_t^{\text{text}}; e_t^{\text{num}}]), \quad e_t = g_t \odot e_t^{\text{text}} + (1 - g_t) \odot e_t^{\text{num}}, \quad (3)$$

where W_g is a trainable parameter matrix, $\sigma(\cdot)$ is the sigmoid function, and e_t denotes the final event embedding. This mechanism enables DyLogNet to dynamically balance the influence of textual semantics and quantitative metrics, ensuring that anomalies manifesting in either form are equally well represented.

3.3 Density-Aware Multi-Relational Graph Construction

A major challenge in log anomaly detection lies in the irregular temporal distribution of events. During peak workloads, thousands of logs may appear within seconds, whereas in stable states, events become sparse. Fixed-size sliding windows often fail under such imbalance: short windows fragment dense bursts, while long windows dilute sparse regions. To overcome this problem, DyLogNet constructs density-aware adaptive graph snapshots with temporal smoothing and bounded adjustment.

The dynamic graph construction module is designed to model both temporal and semantic relationships [20]. Each event is represented as a node, and edges are adaptively defined based on event density and relational dependencies. For a current window of duration T_t containing N_t events, the instantaneous event density is computed as:

$$D_t = \frac{N_t}{T_t}. \quad (4)$$

To avoid instability caused by density fluctuations, DyLogNet maintains an exponentially smoothed density estimate:

$$\hat{D}_t = (1 - \alpha)\hat{D}_{t-1} + \alpha D_t, \quad (5)$$

where $\alpha \in (0, 1)$ controls the temporal smoothing. A smaller α places more weight on the historical estimate $\hat{D}_{t-1}$, yielding a slower but more stable adaptation to density changes, which is beneficial for long-term stationarity. Conversely, a larger α allows the model to react more quickly to sudden bursts or drops in frequency, enhancing short-term responsiveness at the cost of higher variance.

The next window length is then adaptively updated based on the smoothed event density, while simultaneously enforcing both upper and lower temporal bounds to ensure stable and consistent temporal segmentation.

$$T_{t+1} = \text{clip}\left((1 - \beta)T_t + \beta \cdot \frac{\kappa}{\hat{D}_t}, T_{\min}, T_{\max}\right), \quad (6)$$

where $\beta \in (0, 1)$ moderates the update rate, κ controls the target density, and $\text{clip}(\cdot, T_{\min}, T_{\max})$ constrains the window length within the range $[T_{\min}, T_{\max}]$. A small β leads to gradual adjustments of T_t , emphasizing temporal continuity and preventing abrupt window oscillations, while a large β makes the system more agile in adapting to dynamic workloads but can cause transient fluctuations in the window scale. We damp fast oscillations by reusing the last accepted T_t when $|T_{t+1} - T_t|$ falls below a small tolerance ϵ_T , which preserves numerical stability without measurable loss in reactivity.

If no events occur within the current window ($N_t = 0$), the model reverts to a conservative fallback by setting $T_{t+1} = \min(2T_t, T_{\max})$, ensuring temporal continuity without degeneracy. In this case we skip graph construction and gradient updates for snapshot t and only advance the timeline, avoiding empty-graph artifacts.

Each adaptive window defines a graph snapshot $G_t = (V_t, E_t, W_t)$, where V_t denotes nodes, E_t edges, and W_t edge weights. This smoothed adjustment strategy prevents abrupt oscillations in window length and maintains computational stability across dynamic workloads.

Overall, the adaptive construction enables DyLogNet to maintain a balanced temporal resolution, capturing fine-grained transitions during log bursts while preserving sufficient contextual information

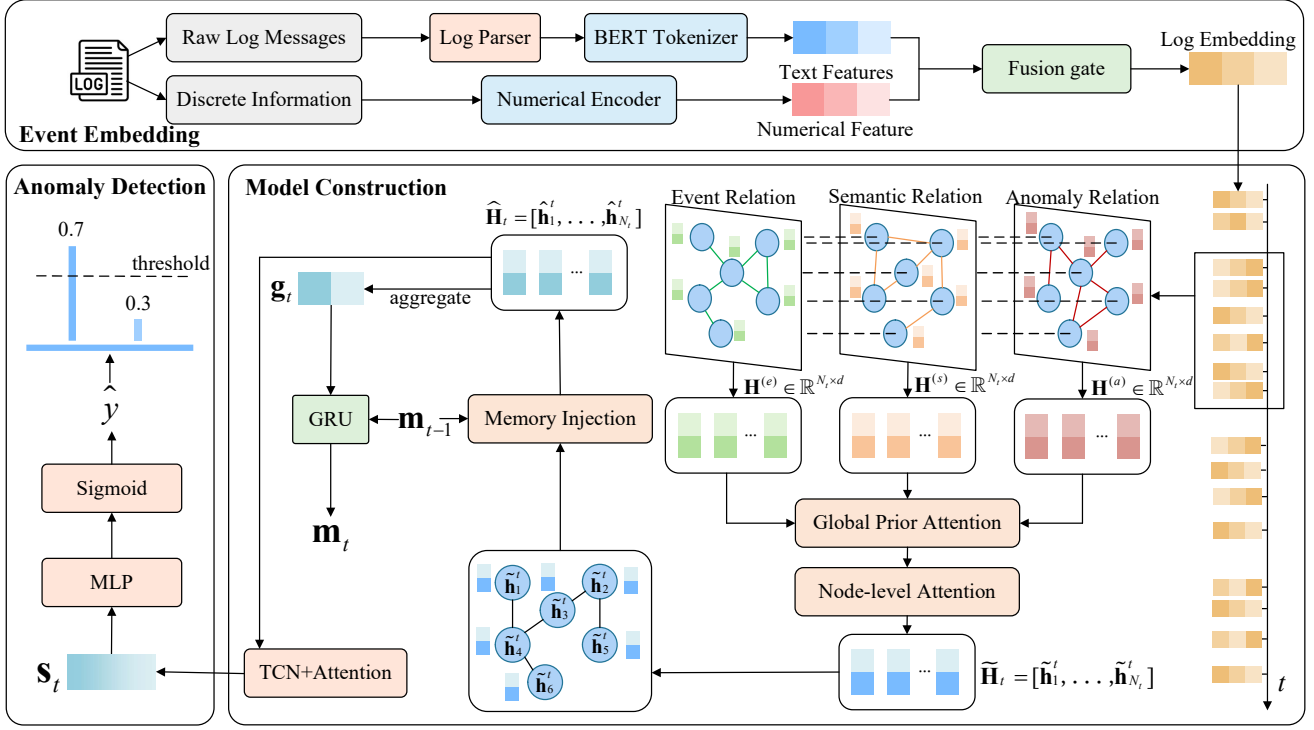


Figure 1: The overall architecture of DyLogNet. Event Embedding fuses semantic and numerical features via a gated mechanism to produce informative log representations. Model Construction builds a dynamic multi-relation graph to model temporal, semantic, and anomaly-aware dependencies. Anomaly Detection employs a detector to infer anomaly scores from the fused event representations.

during low-activity intervals, thus providing a robust foundation for subsequent multi-relational graph modeling.

To comprehensively model different dependency types, DyLogNet constructs three relational layers within each snapshot: Event Relation, Semantic Relation, and Anomaly Relation. Each relation corresponds to a distinct relational perspective and produces its own representation. Formally, for each relation type $r \in \{e, s, a\}$, a graph neural network is applied to its corresponding relation graph to obtain the node embeddings:

$$\mathbf{H}^{(r)} = \text{GNN}(A^{(r)}, E_t), \quad (7)$$

where $A^{(r)}$ is the adjacency matrix of relation r , and $E_t = [e_1, \dots, e_{N_t}]$ denotes the event embeddings within the snapshot. These relation-specific outputs $\mathbf{H}^{(r)}$ are subsequently integrated by the two-level cross-layer attention module.

Event Relation. This relation type focuses on capturing the temporal dependencies between log events [18]. Two events i and j are connected if their timestamps differ by less than a predefined threshold. Accordingly, the edge weight is defined as:

$$w_{ij}^e = \exp(-|\tau_i - \tau_j|/\sigma_t), \quad (8)$$

where τ_i denotes the timestamp of event i , and σ_t controls the temporal decay rate. To avoid dense cliques during bursts, we retain only Top- k temporal neighbors per node.

Semantic Relation. This relation type captures contextual similarity between log events. The edge weight is given by:

$$w_{ij}^s = \frac{e_i \cdot e_j}{\|e_i\| \|e_j\|}, \quad (9)$$

where e_i and e_j are the event embeddings generated by the enhanced semantic encoder. We further apply ℓ_2 -normalization to e_i before computing w_{ij}^s , followed by a cosine-threshold sparsification and adjacency symmetrization, as described above, to avoid constructing overly dense graphs and to enhance numerical stability during training.

Anomaly Relation. The anomaly relation identifies anomaly-prone nodes by comparing normalized embeddings $\hat{e}_i = e_i/\|e_i\|_2$ in a k -nearest neighbor graph. The local anomaly score of node i is:

$$d_i = \frac{1}{k} \sum_{j \in N_k(i)} \text{dist}(\hat{e}_i, \hat{e}_j), \quad (10)$$

where $\text{dist}(\cdot)$ is Euclidean distance. The top- $p\%$ nodes by distance form the anomaly-prone subgraph with edge weights:

$$w_{ij}^a = \exp(-\text{dist}(\hat{e}_i, \hat{e}_j)/\sigma_a), \quad (11)$$

where σ_a is a scaling factor controlling the sensitivity of the exponential decay. A smaller σ_a sharpens the contrast between closely and distantly related nodes, highlighting strong anomaly relationships, while a larger σ_a yields smoother connectivity patterns. Edges

are kept only within the anomaly-prone subgraph to control complexity; cross-subgraph edges are dropped. We also use a projection head $f_{\text{proj}}(\cdot)$ to produce anomaly-aware embeddings $\mathbf{z}_i = f_{\text{proj}}(\mathbf{e}_i)$ for contrastive learning, which increases separability of rare patterns.

3.4 Two-Level Cross-Layer Attention

Different relational layers contribute unevenly to anomaly inference, and simple concatenation overlooks their heterogeneous importance. DyLogNet therefore adopts a two-level cross-layer attention mechanism that captures global relation importance while enabling node-specific adaptation.

Each relation type $r \in \{e, s, a\}$ produces node embeddings $\mathbf{H}^{(r)} = [\mathbf{h}_1^{(r)}, \dots, \mathbf{h}_{N_r}^{(r)}]$. At the global level, DyLogNet learns relation priors α_r to quantify the importance of each relation:

$$\alpha_r = \frac{\exp(a_r / \tau_{\text{rel}})}{\sum_{r'} \exp(a_{r'} / \tau_{\text{rel}})}, \quad (12)$$

where a_r is a trainable logit, and τ_{rel} is a temperature hyperparameter controlling the sharpness of attention.

At the node level, DyLogNet further refines attention by computing node-wise weights:

$$\eta_{i,r} = \log \alpha_r + \mathbf{q}^\top \mathbf{W}_r \mathbf{h}_i^{(r)}, \quad \beta_i^{(r)} = \frac{\exp(\eta_{i,r})}{\sum_{r'} \exp(\eta_{i,r'})}, \quad (13)$$

where $\mathbf{h}_i^{(r)}$ denotes node i 's embedding from relation r , $\mathbf{W}_r$ is a relation-specific transformation, and $\mathbf{q}$ is a shared query vector. The final representation is:

$$\tilde{\mathbf{h}}_i = \sum_{r \in \{e,s,a\}} \beta_i^{(r)} \mathbf{h}_i^{(r)}. \quad (14)$$

This attention process allows DyLogNet to dynamically emphasize the most informative relation for each log event while preserving overall relational consistency across the snapshot.

3.5 Cross-Snapshot Memory Injection

While graph attention captures intra-snapshot relations, anomalies often evolve over time. To retain temporal continuity, DyLogNet introduces a cross-snapshot memory that recurrently propagates information between snapshots.

Formally, let $\tilde{\mathbf{h}}_i^{(t)} \in \mathbb{R}^d$ denote the fused embedding of node i at snapshot t after multi-relational attention aggregation, and let $\mathbf{m}_{t-1} \in \mathbb{R}^d$ represent the global memory summarizing all past snapshots up to time $t-1$. The memory state modulates current node representations through an affine transformation that depends on $\mathbf{m}_{t-1}$:

$$\hat{\mathbf{h}}_i^{(t)} = \gamma_t \odot (\tilde{\mathbf{h}}_i^{(t)} + \mathbf{a}_t) + \beta_t, \quad (15)$$

where $\mathbf{a}_t = \mathbf{W}_a \mathbf{m}_{t-1}$ is an additive bias term, $\gamma_t = \sigma(\mathbf{W}_\gamma \mathbf{m}_{t-1})$ is a gating vector controlling the degree of historical influence, and $\beta_t = \mathbf{W}_\beta \mathbf{m}_{t-1}$ is a shift term that adjusts the baseline of current embeddings. Here, $\mathbf{W}_a, \mathbf{W}_\gamma, \mathbf{W}_\beta \in \mathbb{R}^{d \times d}$ are learnable transformation matrices, and $\sigma(\cdot)$ denotes the element-wise sigmoid function.

The gating operation γ_t acts as a soft controller that determines how much prior knowledge should be injected into the current snapshot, allowing the model to adaptively retain or forget historical signals depending on the context of incoming logs. This

mechanism differs from simple feature concatenation by learning a content-dependent modulation of historical context.

The snapshot-level representation is then computed by mean-pooling over all node embeddings:

$$\mathbf{g}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \hat{\mathbf{h}}_i^{(t)}, \quad (16)$$

where N_t is the total number of log events contained in snapshot t , and $\mathbf{g}_t$ serves as a condensed and informative summary vector representing the current system state. Subsequently, the global memory is recurrently updated using a gated recurrent unit (GRU) to integrate the newly observed information with historical context:

$$\mathbf{m}_t = \text{GRUCell}(\mathbf{g}_t, \mathbf{m}_{t-1}), \quad (17)$$

where $\mathbf{m}_t$ encodes both the current and historical contexts. The recurrent update ensures that temporal dependencies are modeled in a causal and continuous manner.

The cross-snapshot memory injection enables DyLogNet to preserve long-term temporal context and mitigate catastrophic forgetting. By recurrently incorporating prior states into evolving graph representations, it captures slow-evolving anomalies and cross-window contextual shifts that snapshot-only models often miss.

3.6 Anomaly Detection

After memory injection, the resulting sequence of modulated embeddings $\hat{\mathbf{H}}_t = [\hat{\mathbf{h}}_1^{(t)}, \dots, \hat{\mathbf{h}}_{N_t}^{(t)}]$ is further processed by the temporal modeling module, which integrates local and temporal dynamics to obtain a compact snapshot-level representation $\mathbf{s}_t \in \mathbb{R}^d$. Specifically, $\mathbf{s}_t$ is obtained through a sequence of temporal operations:

$$\mathbf{s}_t = \text{AttnPool}(\text{TCN}(\hat{\mathbf{H}}_t + f_{\text{gap}}(\Delta\tau_t))), \quad (18)$$

where $\text{TCN}(\cdot)$ denotes a Temporal Convolutional Network that captures local temporal dependencies through dilated causal convolutions while strictly preserving the chronological order of events. The function $f_{\text{gap}}(\Delta\tau_t)$ encodes the inter-event time gaps into continuous representations, and $\text{AttnPool}(\cdot)$ represents an attention-based pooling operation that aggregates temporal features. This process effectively models short-term temporal dynamics and aggregates the most informative features through learnable attention weights.

A linear prediction head then produces the anomaly probability for snapshot t :

$$\hat{y}_t = \text{sigmoid}(\mathbf{W}_c \mathbf{s}_t + \mathbf{b}_c), \quad (19)$$

where $\mathbf{W}_c$ and $\mathbf{b}_c$ are the classifier parameters. An entire snapshot is labeled anomalous if the probability of the anomaly class exceeds a threshold.

DyLogNet is trained using a hybrid objective that combines supervised classification with contrastive regularization:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \theta \mathcal{L}_{\text{con}}, \quad (20)$$

where λ balances the two terms. The classification loss is the binary cross-entropy:

$$\mathcal{L}_{\text{cls}} = - \sum_t [y_t \log(\hat{y}_t) + (1 - y_t) \log(1 - \hat{y}_t)], \quad (21)$$

Table 1: Statistics of datasets.

Dataset	#Total Logs	#Anomaly	%Anomaly
HDFS	11,175,629	16,838	2.93%
BGL	4,747,963	348,460	7.34%
Thunderbird	10,000,000	4,934	0.05%

and the contrastive InfoNCE loss encourages representation separation between pseudo-labeled normal and anomaly-prone samples:

$$\mathcal{L}_{\text{con}} = \frac{1}{|\mathcal{P}(i)|} \sum_{j \in \mathcal{P}(i)} \log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}, \quad (22)$$

where $\mathcal{P}(i)$ denotes the set of positive samples sharing the same pseudo-label as anchor i , $\text{sim}(\cdot)$ is cosine similarity, and τ is a temperature coefficient. This joint training scheme ensures that DyLogNet not only learns discriminative decision boundaries at the snapshot level but also preserves representation coherence guided by anomaly-aware structural priors.

4 Experiment

We conducted comprehensive comparative and ablation studies on three publicly available log-based anomaly detection datasets to answer the following research questions:

- **RQ1:** Can DyLogNet outperform state-of-the-art log anomaly detection methods in terms of accuracy?
- **RQ2:** How does each component of DyLogNet, including dynamic graph construction, multi-layer graph modeling, and cross-layer fusion, individually contribute to the overall performance of the model?
- **RQ3:** Is DyLogNet effective under challenging scenarios such as few-shot learning and unstable logging environments?

4.1 Experimental Settings

Datasets. We evaluate DyLogNet on three widely used public log-based anomaly detection datasets: HDFS, BGL, and Thunderbird. The HDFS dataset contains logs collected from over 200 Amazon EC2 nodes running Hadoop-based MapReduce jobs. BGL originates from the Blue Gene/L supercomputer, comprising 128,000 processors and over 4.7 million log entries recorded over seven months. Thunderbird is derived from the Thunderbird supercomputer with 9,024 processors and 27 TB of memory; following prior work, we use the earliest 10 million logs for evaluation. A summary of the dataset statistics is provided in Table 1. The evaluation is conducted on the following datasets:

- **HDFS [27]:** A widely-used log dataset containing 11,175,629 log entries corresponding to 575,061 blocks, with 16,838 blocks labeled as anomalous. The anomalies include issues such as hardware faults and data corruption, which are representative of many real-world system errors.
- **BGL [22]:** A large-scale log dataset containing 4,747,963 entries and 348,460 labeled anomalies, spanning diverse failure types, which makes it well-suited for evaluating semantic and contextual modeling methods.

- **Thunderbird:** A log dataset containing 3,000,000 entries with labeled anomalies caused by communication errors and high-frequency patterns, which reflect challenges common in many complex systems.

Metrics. We evaluate the performance of these methods using the widely adopted Precision, Recall and F1-score [33, 36].

Implementation Details. To assess the effectiveness of each method, we initially identified the best hyperparameters with a single seed applied to the validation set. Subsequently, each method was trained with the selected hyperparameters across five different random seeds. All experiments were conducted on a server equipped with an NVIDIA L40S GPU, an Intel Core i9-14900K CPU, 128 GB of RAM, running Ubuntu 20.04. The implementation was developed using Python 3.8.19 and PyTorch 2.0.0. We use the Adam optimizer [13] for parameter optimization with a learning rate of 1×10^{-4} . For Eq. (2), the default value of d is set to 768. For Eq. (5), the default value of α is set to 0.3. For Eq. (6), the default value of β is set to 0.3, κ is set to 20, $T_{\min}$ is set to 10, and $T_{\max}$ is set to 50. For Eq. (7), the default value of σ_t is set to 1. For Eq. (11), the default value of σ_a is set to 1. For Eq. (12), the default value of τ_{rel} is set to 0.8. For Eq. (20), the default value of θ is set to 0.3. For Eq. (22), the default value of τ is set to 0.5.

Baselines. We compare DyLogNet with ten baselines, including (1) **DeepLog** [5] uses LSTM to model system logs as sequences, enabling automatic learning of normal patterns and effective anomaly detection through sequence deviation. (2) **LogAnomaly** [21] combines semantic and quantitative modeling to detect diverse anomalies while reducing false alarms. (3) **PLELog** [28] detects anomalies in unlabeled logs by clustering sequences and estimating labels probabilistically to enable effective supervised training. (4) **LogRobust** [37] addresses log instability by encoding log semantics and leveraging an attention-based Bi-LSTM to detect anomalies in evolving and noisy log sequences. (5) **LLMeLog** [11] enhances log anomaly detection by leveraging large language models to enrich log semantics and fine-tune a BERT-based detector. (6) **KNN** [32] is a lightweight baseline for log anomaly detection that outperforms deep learning methods like NeuralLog in both speed and accuracy on certain datasets. (7) **LogGD** [26] is a graph-based anomaly detection method that transforms log sequences into graphs and uses Graph Transformer Networks to capture structural and semantic patterns for improved detection accuracy and stability. (8) **NeuralLog** [15] is a parsing-free anomaly detection method that uses Transformer-based models to capture the semantics of raw log messages for accurate anomaly identification. (9) **Lograph** [3] constructs a Log-Entity Graph from semantic associations and leverages heterogeneous graph attention to detect anomalies in complex, interleaved logs without predefined grouping.

4.2 Overall Performance (RQ1)

To evaluate the overall effectiveness of the proposed DyLogNet, we conduct comprehensive experiments on three widely used log anomaly detection datasets: HDFS, BGL, and Thunderbird. We compare DyLogNet with a diverse set of baseline methods, including both unsupervised and supervised approaches. Table 2 presents the performance comparison in terms of Precision (P), Recall (R), and

Table 2: Performance Comparison of DyLogNet and Baseline Methods on Three Datasets. The boldface refers to the highest score and the underline indicates the next best result of the models. "*" indicates the statistically significant improvements (i.e., two-sided t-test with $p < 0.05$) over the best baseline.

Category	Model	HDFS			BGL			TDB		
		P	R	F1	P	R	F1	P	R	F1
Unsupervised	DeepLog	0.861	0.951	0.904	0.928	0.894	0.911	0.527	0.795	0.634
	LogAnomaly	0.912	0.982	0.946	0.924	0.865	0.893	0.667	0.835	0.742
	PLELog	0.896	0.978	0.935	0.838	0.853	0.845	0.634	0.804	0.708
Supervised	LogRobust	0.965	0.964	0.964	0.953	0.973	0.962	<u>0.967</u>	0.924	0.945
	NeuralLog	0.921	<u>0.990</u>	0.954	0.976	0.984	0.979	0.862	0.784	0.821
	LogGD	<u>0.975</u>	0.989	<u>0.981</u>	0.981	<u>0.991</u>	<u>0.985</u>	0.856	0.796	0.824
	LLMeLog	0.925	0.975	0.949	0.974	0.993	0.983	0.935	0.913	0.923
	KNN	0.967	0.983	0.974	0.958	0.985	0.971	0.794	0.931	0.850
	Lograph	0.968	0.986	0.976	<u>0.985</u>	0.957	0.970	0.957	<u>0.951</u>	<u>0.953</u>
	DyLogNet	0.985	0.992*	0.988*	0.996*	0.976	0.986*	0.976*	0.962*	0.968*

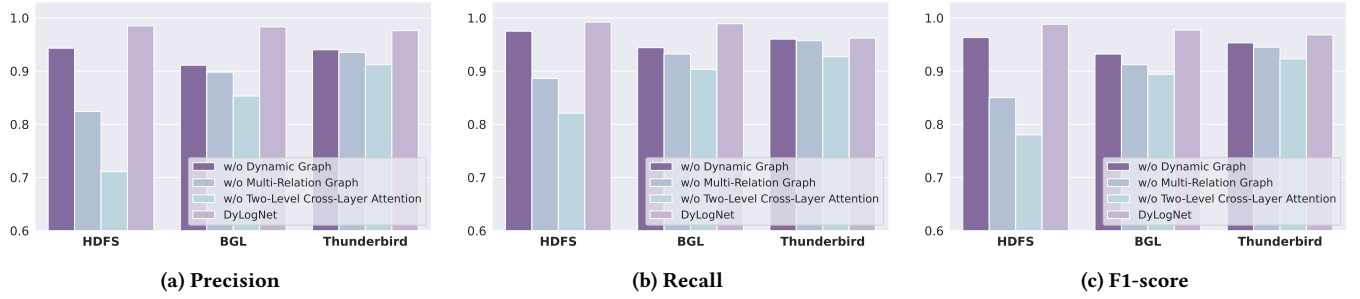


Figure 2: Evaluation results of ablation study.

F1-score (F1) across all datasets. The boldface values indicate the best results, while underlined numbers represent the second best.

Among the unsupervised baselines, LogAnomaly and PLELog achieve competitive results on relatively structured datasets. Specifically, LogAnomaly attains an F1-score of 0.946 on HDFS and 0.893 on BGL, with PLELog following closely behind. These results demonstrate their ability to capture frequent and well-defined anomaly patterns in stable environments. However, both models exhibit a marked performance decline on the more heterogeneous Thunderbird dataset, where the F1-scores drop to 0.742 and 0.708, respectively. This decline highlights limited generalization in complex logs, where irregular timing, interleaving, and overlapping anomalies strain purely sequential models.

Among supervised baselines, LogGD, Lograph, and KNN are consistently strong across datasets. Lograph reaches an F1 of 0.953 on Thunderbird, highlighting the benefit of graph-based dependency modeling, while KNN remains competitive due to its non-parametric robustness to distribution shifts. In contrast, NeuralLog and LogRobust perform well on HDFS and BGL but drop markedly on Thunderbird, likely because their sequential modeling is more sensitive to interleaving and noisy patterns.

In contrast, DyLogNet achieves the best overall performance across all datasets, with F1-scores of 0.988 on HDFS, 0.986 on BGL,

and 0.968 on Thunderbird. These results not only surpass all baselines by a statistically significant margin but also demonstrate DyLogNet’s superior generalization ability across different system architectures and log characteristics. The improvements can be attributed to DyLogNet’s density-aware dynamic graph construction and multi-relational modeling, which jointly capture temporal continuity, semantic coherence, and anomaly-indicative dependencies. Moreover, the cross-snapshot memory injection mechanism enables effective temporal reasoning across log windows, mitigating the limitations of static or sequence-only approaches.

Overall, the results support modeling logs as dynamic multi-relational graphs. Although DyLogNet is not the best in precision in one case, the primary metric is F1-score, which balances precision and recall. Accordingly, DyLogNet achieves the most robust and consistent performance across datasets.

4.3 Ablation Study (RQ2)

To assess the contribution of DyLogNet’s components, we perform ablations on three representative datasets, focusing on dynamic graph construction, multi-relation modeling, and cross-layer attention fusion. Each variant removes or simplifies the target module while keeping the remainder unchanged. The resulting F1-scores are reported in Figure 2.

Table 3: Evaluation results of few-shot scenario.

Model	Metric	BGL			Thunderbird		
		5%	10%	15%	5%	10%	15%
DeepLog	P	<u>0.821</u>	0.837	0.827	0.165	0.238	0.473
	R	0.951	0.896	0.934	0.341	0.483	0.711
	F1	0.881	0.865	0.877	0.222	0.318	0.568
LogAnomaly	P	0.829	<u>0.846</u>	0.833	<u>0.607</u>	0.594	0.618
	R	0.911	<u>0.935</u>	0.926	<u>0.643</u>	0.817	<u>0.784</u>
	F1	0.868	<u>0.888</u>	0.877	<u>0.624</u>	0.687	0.691
PLELog	P	0.053	0.768	0.819	0.539	<u>0.681</u>	0.664
	R	0.215	0.841	0.906	0.597	0.713	0.753
	F1	0.085	0.802	0.860	0.567	<u>0.696</u>	0.705
LogRobust	P	0.058	0.176	0.647	0.101	0.496	0.617
	R	0.147	0.468	0.793	0.054	0.571	0.680
	F1	0.083	0.255	0.712	0.070	0.530	0.646
NeuralLog	P	0.044	0.607	0.671	0.039	0.514	0.307
	R	0.581	0.678	0.695	0.140	0.062	0.241
	F1	0.081	0.640	0.682	0.061	0.110	0.270
Lograph	P	0.796	0.815	<u>0.892</u>	0.281	0.495	<u>0.763</u>
	R	0.868	0.879	0.913	0.076	0.257	<u>0.697</u>
	F1	0.830	0.845	<u>0.902</u>	0.119	0.338	<u>0.728</u>
DyLogNet	P	0.811	0.864	0.916	0.617	0.783	0.874
	R	<u>0.937</u>	0.960	0.957	0.780	<u>0.741</u>	0.918
	F1	<u>0.869</u>	0.909	0.936	0.689	0.761	0.895

First, w/o Dynamic Graph replaces the density-aware construction with a fixed static window, removing adaptation to log-density variation and temporal granularity, and weakening the modeling of dynamic behaviors. Second, w/o Multi-Relation Graph retains only one relation and discards semantic and anomaly-propensity links, limiting the model to a single structural view. Third, w/o Two-Level Cross-Layer Attention replaces hierarchical attention fusion with uniform averaging across relations, eliminating adaptive weighting and neglecting heterogeneity in relational importance.

Ablation results show that all components are essential to DyLogNet. On HDFS, removing the dynamic graph module reduces F1 from 0.988 to 0.963, indicating its importance for evolving temporal dependencies. Further removing multi-relation modeling drops F1 to 0.85, validating the benefit of relational diversity, while excluding cross-layer attention causes the largest decline to 0.78, highlighting the role of adaptive feature fusion in robust detection. Across BGL and Thunderbird, DyLogNet consistently outperforms all ablations, confirming the complementary value of dynamic graph construction, multi-relation modeling, and cross-layer attention, with each component contributing synergistically to anomaly detection.

4.4 Few-shot Scenario (RQ3)

To assess DyLogNet under limited supervision, we adopt a few-shot setting that reflects real deployments where labeled anomaly logs are scarce. This scenario is practically important because large-scale log annotation is time-consuming and costly [17, 19]. Therefore, we aim to investigate whether DyLogNet can maintain robust and competitive performance when trained with only a small subset of labeled samples.

We conduct experiments on the BGL and Thunderbird datasets by constraining the proportion of labeled training data to 5%, 10%, and 15% of the original training set, while keeping the test set fixed. Under each setting, DyLogNet is compared with several state-of-the-art baseline methods, including both sequence-based and graph-based models, to comprehensively evaluate its adaptability to few-shot conditions.

As presented in Table 3, DyLogNet consistently achieves superior results across all settings. On the BGL dataset, the model attains an F1-score of 0.869 with only 5% labeled data, and the score further increases to 0.936 when 15% of the data is used. Compared to the strongest baselines, LogAnomaly and Lograph, which achieve 0.868 and 0.902 F1-scores under the same settings, DyLogNet demonstrates clear advantages in both precision and recall. Particularly, its recall remains as high as 0.937 under extreme data scarcity, suggesting strong robustness in detecting rare anomaly patterns.

On Thunderbird, DyLogNet shows a clear advantage in few-shot settings. With 5% labeled data, it attains an F1-score of 0.689, outperforming all baselines. As supervision increases, performance improves steadily, reaching 0.895 at 15%, surpassing the best baseline by over 0.2 on average. It also maintains balanced precision and recall, indicating strong sensitivity and specificity in anomaly detection.

Overall, the results validate DyLogNet’s data efficiency and generalization. With density-aware dynamic graph construction and cross-snapshot reasoning, it captures temporal, semantic, and structural dependencies under limited supervision. The strong few-shot performance highlights its practical potential for deployments where labeled anomalies are scarce and annotation costs are high.

5 Conclusion

We propose DyLogNet, a dynamic multi-relational graph framework for log anomaly detection. It captures evolving dependencies via three relations: Event for temporal proximity, Semantic for contextual similarity, and Anomaly for contrastive anomaly enhancement. DyLogNet further employs density-aware dynamic graph construction and GRU-based recurrent fusion to handle irregular logs and preserve long-term continuity across snapshots. Experiments on three public datasets show consistent gains in accuracy, robustness, and generalization, particularly in few-shot settings.

Future Work. In future research, we plan to extend DyLogNet for low-latency deployment in large-scale distributed systems by optimizing its dynamic graph update mechanism and reducing the computational complexity of GNN layers. We also intend to explore adaptive streaming extensions of DyLogNet for online anomaly detection in high-throughput environments, enabling real-time system monitoring and rapid fault diagnosis.

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References

- [1] Yang Cao, Haolong Xiang, Hang Zhang, Ye Zhu, and Kai Ming Ting. 2025. Anomaly detection based on isolation mechanisms: A survey. *Machine Intelligence Research* (2025), 1–17.
- [2] Zeming Chen, Qiyue Gao, and Lawrence S Moss. 2021. NeuralLog: Natural Language Inference with Joint Neural and Logical Reasoning. In *The 10th Conference on Lexical and Computational Semantics*. 78.
- [3] Guojun Chu, Jingyu Wang, Qi Qi, Haifeng Sun, Zirui Zhuang, Bo He, Yuhao Jing, Lei Zhang, and Jianxin Liao. 2025. Anomaly Detection on Interleaved Log Data With Semantic Association Mining on Log-Entity Graph. *IEEE Transactions on Software Engineering (TSE)* (2025).
- [4] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies*. 4171–4186.
- [5] Min Du, Feifei Li, Guineng Zheng, and Vivek Srikumar. 2017. Deeplog: Anomaly detection and diagnosis from system logs through deep learning. In *Proceedings of the 2017 ACM SIGSAC conference on computer and communications security*. 1285–1298.
- [6] Mengwei Fan, Xiuguo Zhang, Peipeng Wang, and Zhiying Cao. 2025. Multi-modal anomaly detection for microservice system through nested graph diffusion reconstruction. *Applied Intelligence* 55, 11 (2025), 784.
- [7] Kexiong Fei, Jiang Zhou, Yucan Zhou, Xiaoyan Gu, Haihui Fan, Bo Li, Weiping Wang, and Yong Chen. 2025. LaAeb: A comprehensive log-text analysis based approach for insider threat detection. *Computers & Security* 148 (2025), 104126.
- [8] Hongcheng Guo, Jian Yang, Jiaheng Liu, Jiaqi Bai, Boyang Wang, Zhoujun Li, Tieqiao Zheng, Bo Zhang, Junran Peng, and Qi Tian. 2024. Logformer: A pre-train and tuning pipeline for log anomaly detection. In *Proceedings of the AAAI conference on artificial intelligence (AAAI)*, Vol. 38. 135–143.
- [9] Haixuan Guo, Shuhan Yuan, and Xintao Wu. 2021. Logbert: Log anomaly detection via bert. In *2021 international joint conference on neural networks (IJCNN)*. IEEE, 1–8.
- [10] Jie Guo, Meiting Wang, Yan Zhou, Bin Song, Yuhao Chi, Wei Fan, and Jianglong Chang. 2023. HGAN: Hierarchical graph alignment network for image-text retrieval. *IEEE Transactions on Multimedia* 25 (2023), 9189–9202.
- [11] Minghua He, Tong Jia, Chiming Duan, Huaqian Cai, Ying Li, and Gang Huang. 2024. Llmelog: An approach for anomaly detection based on llm-enriched log events. In *2024 IEEE 35th International Symposium on Software Reliability Engineering (ISSRE)*. IEEE, 132–143.
- [12] Tong Jia, Ying Li, Yong Yang, Gang Huang, and Zhonghai Wu. 2022. Augmenting log-based anomaly detection models to reduce false anomalies with human feedback. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. 3081–3089.
- [13] Diederik P Kingma and Jimmy Ba. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014).
- [14] Max Landauer, Florian Skopik, and Markus Wurzenberger. 2024. A critical review of common log data sets used for evaluation of sequence-based anomaly detection techniques. *Proceedings of the ACM on Software Engineering* 1, FSE (2024), 1354–1375.
- [15] Van-Hoang Le and Hongyu Zhang. 2021. Log-based anomaly detection without log parsing. In *2021 36th IEEE/ACM International Conference on Automated Software Engineering (ASE)*. IEEE, 492–504.
- [16] Van-Hoang Le and Hongyu Zhang. 2022. Log-based Anomaly Detection with Deep Learning: How Far Are We?. In *2022 IEEE/ACM 44th International Conference on Software Engineering (ICSE)*. IEEE, 1356–1367.
- [17] Van-Hoang Le and Hongyu Zhang. 2023. Log parsing with prompt-based few-shot learning. In *2023 IEEE/ACM 45th International Conference on Software Engineering (ICSE)*. IEEE, 2438–2449.
- [18] Zhong Li, Jiayang Shi, and Matthijs Van Leeuwen. 2024. Graph neural networks based log anomaly detection and explanation. In *Proceedings of the 2024 IEEE/ACM 46th international conference on software engineering: companion proceedings*. 306–307.
- [19] Yilun Liu, Shimin Tao, Weibin Meng, Feiyu Yao, Xiaofeng Zhao, and Hao Yang. 2024. Logprompt: Prompt engineering towards zero-shot and interpretable log analysis. In *Proceedings of the 2024 IEEE/ACM 46th International Conference on Software Engineering: Companion Proceedings*. 364–365.
- [20] Franco Manessi, Alessandro Rozza, and Mario Manzo. 2020. Dynamic graph convolutional networks. *Pattern Recognition* 97 (2020), 107000.
- [21] Weibin Meng, Ying Liu, Yichen Zhu, Shenglin Zhang, Dan Pei, Yuqing Liu, Yihao Chen, Ruizhi Zhang, Shimin Tao, Pei Sun, et al. 2019. Loganomaly: Unsupervised detection of sequential and quantitative anomalies in unstructured logs.. In *IJCAI*, Vol. 19. 4739–4745.
- [22] Adam Oliner and Jon Stearley. 2007. What supercomputers say: A study of five system logs. In *37th annual IEEE/IFIP international conference on dependable systems and networks (DSN'07)*. IEEE, 575–584.
- [23] Zumin Wang, Jiyu Tian, Hui Fang, Liming Chen, and Jing Qin. 2022. LightLog: A lightweight temporal convolutional network for log anomaly detection on the edge. *Computer Networks* 203 (2022), 108616.
- [24] Haolong Xiang, Xuyun Zhang, Hongsheng Hu, Lianyong Qi, Wanchun Dou, Mark Dras, Amin Beheshti, and Xiaolong Xu. 2023. OptIForest: optimal isolation forest for anomaly detection. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*. 2379–2387.
- [25] Haolong Xiang, Xuyun Zhang, Xiaolong Xu, Amin Beheshti, Lianyong Qi, Yujie Hong, and Wanchun Dou. 2024. Federated learning-based anomaly detection with isolation forest in the iot-edge continuum. *ACM Transactions on Multimedia Computing, Communications and Applications* (2024).
- [26] Yongzheng Xie, Hongyu Zhang, and Muhammad Ali Babar. 2022. Loggd: Detecting anomalies from system logs with graph neural networks. In *2022 IEEE 22nd International conference on software quality, reliability and security (QRS)*. IEEE, 299–310.
- [27] Wei Xu, Ling Huang, Armando Fox, David Patterson, and Michael Jordan. 2009. Largescale system problem detection by mining console logs. In *Proceedings of SOSP*, Vol. 9. 1–17.
- [28] Lin Yang, Junjie Chen, Zan Wang, Weijing Wang, Jiajun Jiang, Xuyuan Dong, and Wenbin Zhang. 2021. Pelog: Semi-supervised log-based anomaly detection via probabilistic label estimation. In *2021 IEEE/ACM 43rd International Conference on Software Engineering: Companion Proceedings (ICSE-Companion)*. IEEE, 230–231.
- [29] Xixuan Yang, Xin Huang, Chiming Duan, Tong Jia, Shandong Dong, Ying Li, and Gang Huang. 2025. Enhancing web service anomaly detection via fine-grained multi-modal association and frequency domain analysis. In *Companion Proceedings of the ACM on Web Conference 2025*. 548–556.
- [30] Ali Yilmaz and Resul Das. 2025. A novel hybrid approach combining GCN and GAT for effective anomaly detection from firewall logs in campus networks. *Computer Networks* 259 (2025), 111082.
- [31] Shi Ying, Bingming Wang, Lu Wang, Qingshan Li, Yishi Zhao, Jianga Shang, Hao Huang, Guoli Cheng, Zhe Yang, and Jiangyi Geng. 2021. An improved KNN-based efficient log anomaly detection method with automatically labeled samples. *ACM Transactions on Knowledge Discovery from Data (TKDD)* 15, 3 (2021), 1–22.
- [32] Boxi Yu, Jiayi Yao, Qiwei Fu, Zhiqing Zhong, Haotian Xie, Yaoliang Wu, Yuchi Ma, and Pinjia He. 2024. Deep learning or classical machine learning? an empirical study on log-based anomaly detection. In *Proceedings of the 46th IEEE/ACM international conference on software engineering*. 1–13.
- [33] Chenyangguang Zhang, Tong Jia, Guopeng Shen, Pinyan Zhu, and Ying Li. 2024. Metalog: Generalizable cross-system anomaly detection from logs with meta-learning. In *Proceedings of the IEEE/ACM 46th International Conference on Software Engineering*. 1–12.
- [34] Lingzhe Zhang, Tong Jia, Mengxi Jia, Ying Li, Yong Yang, and Zhonghai Wu. 2024. Multivariate log-based anomaly detection for distributed database. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. 4256–4267.
- [35] Lingzhe Zhang, Tong Jia, Mengxi Jia, Yifan Wu, Hongyi Liu, and Ying Li. 2025. ScalaLog: Scalable Log-Based Failure Diagnosis Using LLM. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 1–5.
- [36] Lingzhe Zhang, Tong Jia, Kangjin Wang, Mengxi Jia, Yong Yang, and Ying Li. 2024. Reducing events to augment log-based anomaly detection models: An empirical study. In *Proceedings of the 18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement*. 538–548.
- [37] Xu Zhang, Yong Xu, Qingwei Lin, Bo Qiao, Hongyu Zhang, Yingnong Dang, Chunyu Xie, Xinsheng Yang, Qian Cheng, Ze Li, et al. 2019. Robust log-based anomaly detection on unstable log data. In *Proceedings of the 2019 27th ACM joint meeting on European software engineering conference and symposium on the foundations of software engineering*. 807–817.
- [38] Yifan Zhang, Haolong Xiang, Xiaolong Xu, Zishun Rui, Xiaoyong Li, Lianyong Qi, and Fei Dai. 2025. MEGAD: a memory-efficient framework for large-scale attributed graph anomaly detection. In *Proceedings of the Thirty-fourth International Joint Conference on Artificial Intelligence*. 3651–3659.
- [39] Yifan Zhang, Haolong Xiang, Xiaolong Xu, Zishun Rui, Xiaoyong Li, Lianyong Qi, and Fei Dai. 2025. MEGAD: A Memory-Efficient Framework for Large-Scale Attributed Graph Anomaly Detection. In *Proceedings of the Thirty-fourth International Joint Conference on Artificial Intelligence (IJCAI)*.